PROJECT TITLE

School of Computing

Bachelor of Technology
in
Computer Science & Engineering

By

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Full Stack Application Development
Slot : E2-B20

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN Dr.SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY
(Deemed to be University Estd u/s 3 of UGC Act, 1956)
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May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled ”EventVerse: Smart College Event Hub” by Vaishnavi Vennapusa (VTU26447) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and ref- erenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke pe- nal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

(Vennapusa Vaishnavi)

Date:06/05/2026

ABSTRACT

The Smart Campus Event Platform is a web-based application developed to simplify and automate event management in educational institutions. Traditional event handling methods involve manual processes, which are time-consuming and prone to errors. This system provides a centralized solution for managing all event-related activities efficiently.The platform allows administrators to create and manage events, while students can easily browse and register for them. It includes features such as secure user authentication, digital attendance tracking, automated certificate generation, and email notifications. The system is developed using modern technologies like React.js for the frontend and PHP[2] with MySQL[1] for the backend.By automating key processes, the system reduces manual work, improves accuracy, and enhances user experience. The platform ensures better communication between administrators and participants and provides a reliable and scalable solution for campus event management.

Keywords: Smart Campus, Event Management System, Web Application, React.js, PHP, MySQL, Automation, Attendance Tracking, Certificate Generation, JWT

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LIST OF ACRONYMS AND ABBREVIATIONS

DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	HyperText Transfer Protocol
SMTP	Simple Mail Transfer Protocol
CRUD	Create, Read, Update, Delete
XAMPP	Cross-Platform, Apache, MySQL, PHP, Perl
URL	Uniform Resource Locator
REST	Representational State Transfer

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Chapter 1

INTRODUCTION

1.1 Introduction

The Smart Campus Event Platform is a modern web-based system developed to simplify and automate the management of events within educational institutions. In many campuses, event-related activities such as announcements, registrations, attendance tracking, and certificate distribution are still handled manually leading to inefficiencies and errors.

This platform provides a centralized solution where administrators can create and manage events, and students can easily discover and participate in them. With features like secure authentication, real-time attendance tracking, and digital certificate generation, the platform enhances both usability and efficiency.By leveraging modern web technologies, the system delivers a responsive and user-friendly experience, making event management more organized, transparent,accessible.

1.2 Aim of the Project

The main aim of the Smart Campus Event Platform is to develop an efficient and user-friendly system that automates the entire lifecycle of campus events. To provide a centralized platform for managing campus events To simplify event discovery and registration for students To ensure secure authentication and data handling To automate attendance tracking and reduce manual errors To generate and distribute certificates digitally To improve communication through notifications and email services

1.3 Scope of the Project

The scope of the Smart Campus Event Platform is focused on improving event management processes within colleges and universi- ties. The system is designed to be used by administrators and stu- dents within a campus environment. It covers functionalities such as event creation, registration, participation tracking, and certification. The platform can handle multiple events simultaneously and supports different types of users with role-based access. Although the current system is designed for campus use, it can be extended in the future to support larger organizations, integrate mobile applications, include advanced analytics, and support third-party integrations. The project provides a scalable foundation for future enhancements.

1.4 Framework

The framework of the Smart Campus Event Platform follows a client-server architecture. The frontend is developed using React.js, which provides a dynamic and responsive user interface. The backend is implemented using PHP[1], which handles server-side logic and communication with the database. MySQL[2] is used as the database management system to store and retrieve data efficiently.

The system uses RESTful APIs for communication between the frontend and backend. JSON[1] Web Tokens (JWT) are used for secure authentication and authorization. Additional libraries and APIs such as SendGrid are used for email notifications, and FPDF is used for generating certificates in PDF format.

The platform is deployed using the XAMPP[2] server environment, which includes Apache and MySQL, providing a complete local development and testing setup. This framework ensures modularity, scalability, and efficient performance of the system.

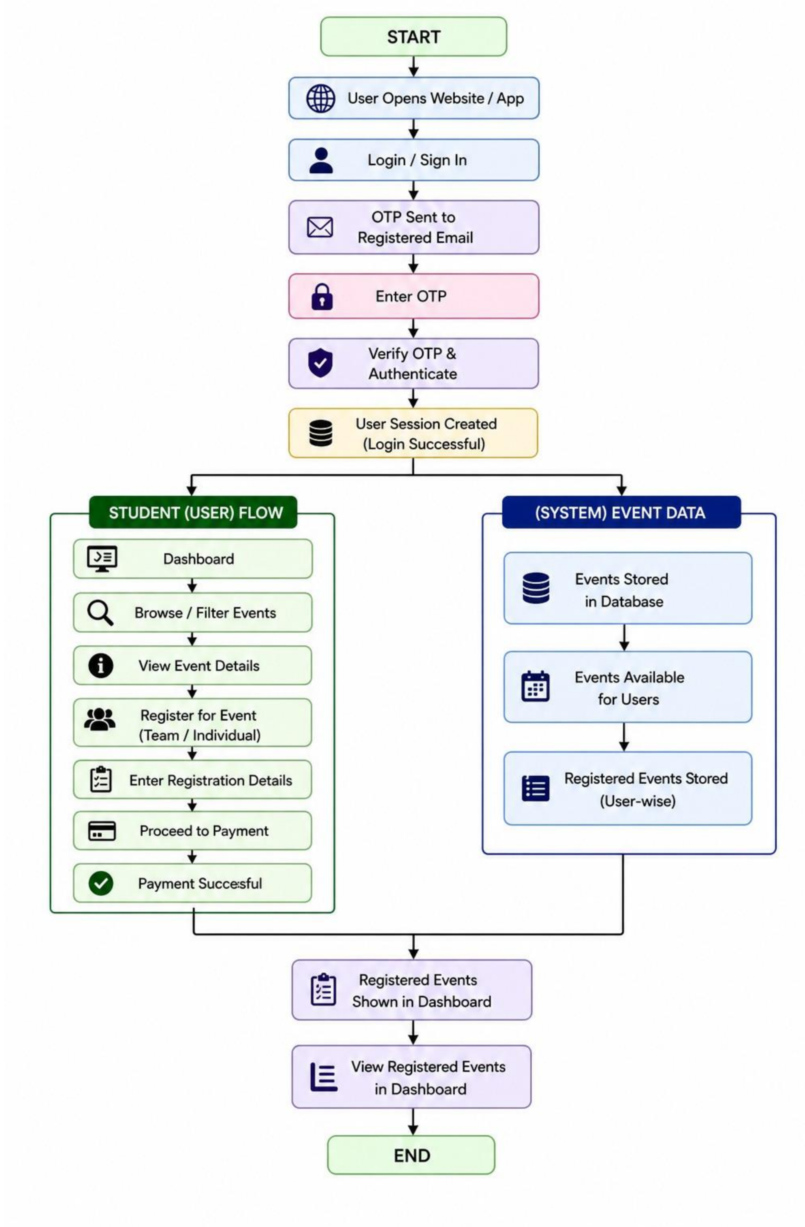

Figure 1.1: Flow Diagram

The Figure 1.1 illustrates the workflow of the Nexora Smart EventManagement System. It begins with the user accessing the website,followed by login or signup and authentication. Based on the user role,the system directs either to the student flow or admin flow.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Various event management applications are available in the market that provide comprehensive solutions for organizing, managing, and promoting events. These platforms help automate tasks such as event creation, registration, communication, attendance tracking, and reporting.

One of the widely used platforms is Cvent, which offers an end-to-end solution including event registration, marketing, venue management, and analytics. It is mainly used for large-scale and enterprise-level events due to its advanced features and integrations.

Another popular platform is Bizzabo, which focuses on delivering a complete event experience through customizable tools, mobile applications, and attendee engagement features like live polls and notifications. vFairs is a modern event management system that supports both physical and virtual events. It provides features such as event registration, email campaigns, user management, QR-based check-in, and networking tools, making it suitable for hybrid events.

In the education sector, platforms like Creatrix Campus Event Management System are specifically designed for colleges and universities. These systems offer centralized dashboards, automated scheduling, registration management, and communication tools tailored for campus environments. Additionally, tools like Eventwise focus on budgeting, ticket tracking, and multi-event reporting, helping organizations manage multiple events efficiently in one platform. From the survey, it is observed that most existing applications provide powerful features such as registration management, communication tools, analytics, and event tracking. However, many systems are either complex, costly, or not specifically tailored for campus-level usage.

The proposed Smart Campus Event Platform aims to overcome these limitations by providing a simple, user-friendly, and cost-effective solution specifically designed for educational institutions, with focused features like attendance tracking, automated certification, and easy event discovery.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

The frontend of the Smart College Event Platform is developed using React.js with Vite as the build tool. It provides a fast, responsive, and interactive user interface for students, organizers, and administrators. Thefrontend handles user interactions such as registration, login, event browsing, and certificate access, ensuring a smooth and user-friendly experience.

3.1.1 Environment Setup

The development environment requires the following tools:

- XAMPPServer
- Apache and MySQL services
- PHPsupport
- phpMyAdmin for database management

The setup procedure is as follows:

1. Install XAMPP.

2. Start Apache and MySQL services.

3. Place backend files inside the htdocs folder.

4. Configure the database connection in the PHP files.

3.1.2 Structure of the Project

The backend folder structure consists of the following components:

• **api**/– Contains API endpoints

• **config**/– Database configuration files

• **auth**/– Handles login and JWT authentication

• **events**/– Manages event creation and updates

• **certificates**/– Generates PDF certificates

• **notifications**/– Handles email services

3.1.3 Execution Procedure

1. Start Apache and MySQL services in XAMPP.

2. Import the database using phpMyAdmin.

3. Run the backend using localhost.

4. Connect the frontend application with backend APIs

3.2 Backend Implementation

The backend of the Smart College Event Platform is developed using PHP and MySQL. It is responsible for handling business logic, processing user requests, managing authentication, storing data, and connecting the frontend with the database. The backend ensures se cure communication between users and the system while maintaining smooth performance

3.2.1 Environment Setup

To set up the backend environment, XAMPP server is used because it provides Apache, PHP, and MySQL in one package. After installing XAMPP, Apache and MySQL services must be started from the con trol panel. The project backend files are placed inside the htdocs folder. Acode editor such as Visual Studio Code can be used for writing and editing PHP files. phpMyAdmin is used for database creation and management

3.2.2 Structure of the Project

The backend project is organized into multiple folders and files for better maintenance and development. Configuration files are used for database connection settings. API files handle requests from the fron tend. Authentication files manage login, registration, and JWT token validation. Event management files perform create, update, delete, and view operations. Notification files handle email sending, while certifi cate files generate PDF certificates. This modular structure improves readability and scalability of the application

3.2.3 Execution Procedure

To execute the backend, first start Apache and MySQL services in

XAMPP. Then open phpMyAdmin and import the project database file. After that, place the backend project folder inside htdocs. Open a browser and run the project using localhost. The frontend communi cates with backend APIs through HTTP requests. Once the connection is established, users can log in, manage events, and perform other op erations successfully.

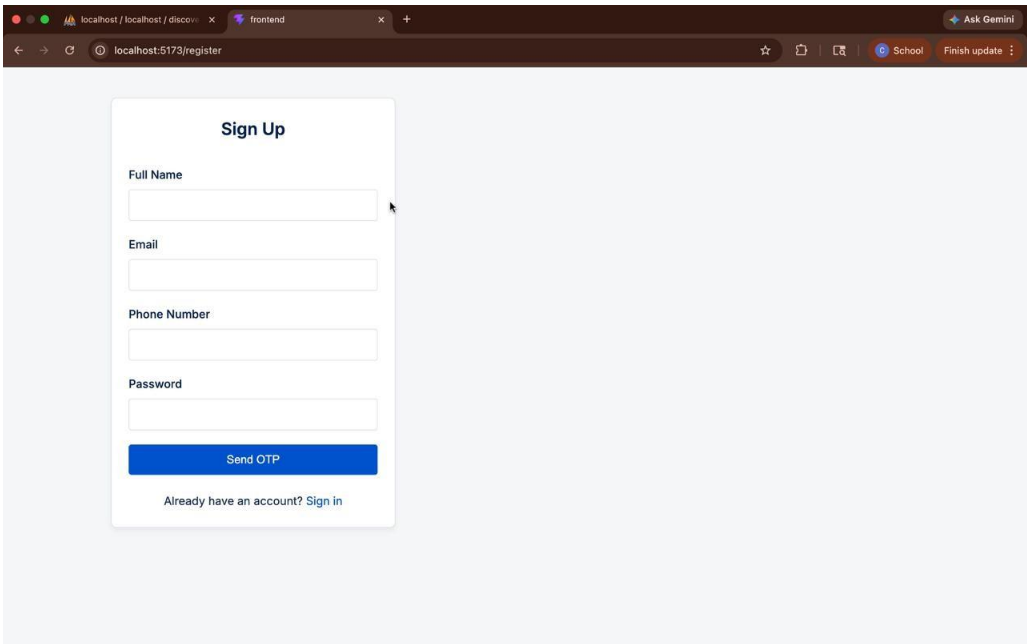

Figure 3.1:StudentSignUp

The Sign Up page allows new users to create an account by entering details such as full name, email, phone number, and password. The form is simple and easy to understand, ensuring smooth user interaction. An OTP verification feature is included to improve security, where users receive a code to verify their details. Additionally, a “Sign in” option is provided for existing users, allowing easy navigation between registration and login.

3.3 Database Implementation

The Smart Campus Event Platform uses MySQL as the database, managed through XAMPP. The backend (PHP) connects using PDO for secure data handling.

The database follows a relational structure with tables like users, events, registrations, attendance, and certificates, connected using keys.

Prepared statements are used to ensure security and prevent SQL injection. The design ensures efficient data storage, easy retrieval, and reliable system performance.

3.3.1 Database Configuration

The database for the Smart Campus Event Platform is implemented using MySQL. The database is created and managed through phpMyAdmin in the XAMPP environment. A connection between the backend (PHP) and the database is established using PDO (PHP Data Objects), which ensures secure and efficient data handling.

The configuration includes defining database name, username, password, and host details in a configuration file. Proper error handling and secure queries (prepared statements) are used to prevent SQL injection and ensure data integrity.

3.1.2 Schema Structure

The database follows a relational schema design. It consists of multiple tables connected through primary and foreign keys to maintain relationships between data.

Key entities in the schema include:

Users – stores user details (admin and students)

Events – stores event information

Registrations – links users with events

Attendance – records participation status

Certificates – stores generated certificate details

3.1.3 Tables created

Several tables are created for the proper functioning of the system.The users table stores student and admin information. The events table stores event title, description, date, and venue. The registrations table stores student participation details. The attendance table stores check in records. The certificates table stores generated certificate information. Notification logs may also be stored for tracking sent messages. These tables together support all operations of the platform

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Smart Campus Event Platform follows a client-server architecture. The frontend (client side) is developed using React.js, which interacts with the backend through RESTful APIs. The backend is implemented using PHP, which processes requests, applies business logic, and communicates with the MySQL database. The system uses JWT (JSON Web Tokens) for secure authentication and authorization. The architecture ensures separation between presentation, logic, and data layers, making the system scalable, secure, and easy to maintain.

4.2 DataFlow Diagram

The data flow in the system describes how information moves between users and system components. Users (Admin/Student) interact with the frontend interface Requests are sent to the backend through

APIs Backend processes the request and interacts with the database The database returns the required data Backend sends the response back to the frontend The frontend displays the result to the user

4.3 Module 1:

This module is responsible for managing user authentication and ensuring data security within the system. It allows new users to register by providing necessary details and enables existing users to log in securely using their credentials. Passwords are encrypted using secure hashing techniques to prevent unauthorized access. The module uses JWT (JSON Web Token) for session management, allowing users to stay authenticated across requests without storing session data on the server. It also implements role-based access control, distinguishing between administrators and students, ensuring that only authorized users can access specific functionalities. Additional validation mechanisms are included to protect against invalid inputs and unauthorized login attempts.

4.4 Module 2:

This module is designed for administrators to efficiently manage all event-related activities. It provides functionalities to create new events by entering details such as event name, description, date, time, venue, and participant limits. Administrators can update event information whenever required and also have the ability to delete events that are no longer needed. The module ensures proper scheduling and organization of events, avoiding conflicts and maintaining consistency. It may also include features for tracking the number of registered participants and monitoring event status. This centralized control helps in maintaining smooth coordination and effective management of campus events.

4.5 Module 3:

This module focuses on providing a user-friendly interface for students to explore and participate in events. Students can view a list of available events along with detailed information such as event description, schedule, and participation criteria. The module includes search and filtering options, allowing users to easily find events based on their interests or categories. Students can register for events individually or as a team, depending on the event type. It ensures a smooth registration process with proper validation and confirmation. The module may also display the list of registered events in the user dashboard, helping students keep track of their participation.

Advantages of this Project

The Smart Campus Event Platform offers several benefits

a) Centralized Management

All event-related activities are handled in a single system, reducing confusion and improving efficiency.

b) Automation:

Automates registration, attendance tracking, and certificate generation, minimizing manual work and errors.

c) User-Friendly Interface

Provides an easy-to-use and responsive interface for both administrators and students.

Disadvantages

Despite its benefits, the system has some limitations:

Requires internet connectivity for access

Initial setup and configuration may take time

Limited to campus-level usage unless further expanded

Chapter 5

RESULTS AND DISCUSSIONS

The Smart Campus Event Platform works successfully with all main features. Users can register, log in, and access the system securely. Admins can create and manage events easily. Students can view and register for events without issues.

Attendance is tracked digitally, reducing errors. Certificates are generated automatically and sent via email. The system saves time, reduces manual work, and improves efficiency. It is user-friendly and reliable. However, it runs on a local server and can be improved with cloud deployment and more features. Overall, the system is effective for managing campus events.

Table 5.1: Comparison Between Existing and Proposed System

Criteria	**Existing System**	**Proposed System**
Registration	Manual or separate platforms	Online centralized registration
Event Management	Paper-based / scattered tools	Single dashboard for all events
Attendance	Manual marking	Digital attendance tracking
Certificates	Manually created and distributed	Auto PDF generation & email delivery
Notifications	Notice boards / emails manually	Automated email notifications
Time Efficiency	Time-consuming	Fast and automated processing
Errors	High chances of human error	Reduced errors through automation

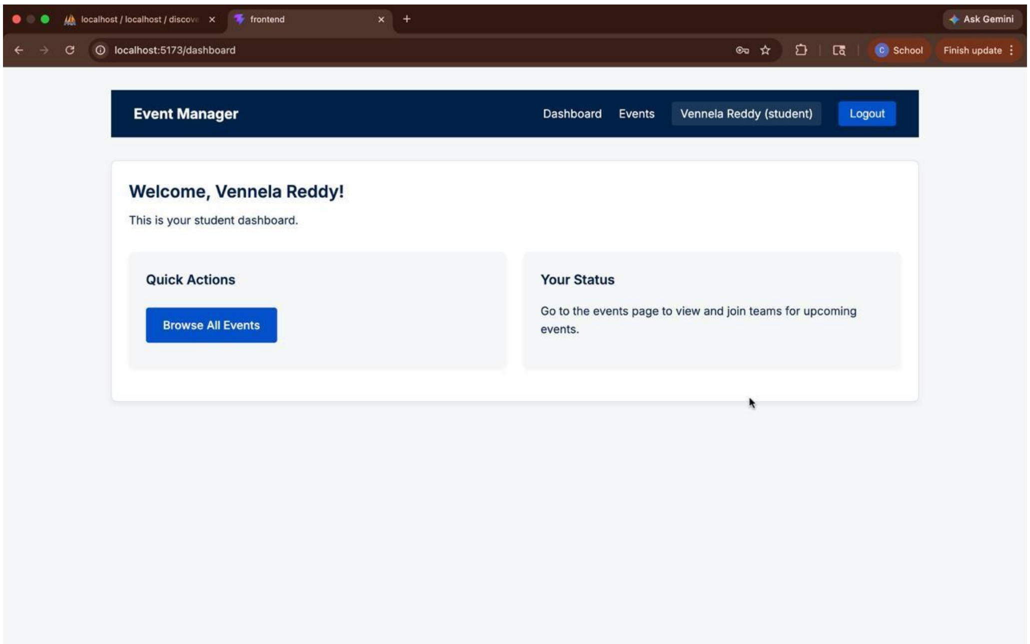

Figure 5.1: S t u d e n t D a s h b o a r d

The dashboard page provides a personalized interface for users after successful login. It displays a welcome message along with the user’s name, creating a customized experience. The navigation bar includes options such as Dashboard, Events, and Logout for easy access to different features. The page contains sections like “Quick Actions,” which allows users to easily browse avail- able events, and “Your Status,” which provides information about participating in upcoming events. The layout is simple, organized, and user-friendly, enabling users to navigate and perform actions efficiently. Overall, the dashboard acts as a central hub where users can manage and access all event- related activities in a clear and convenient manner.

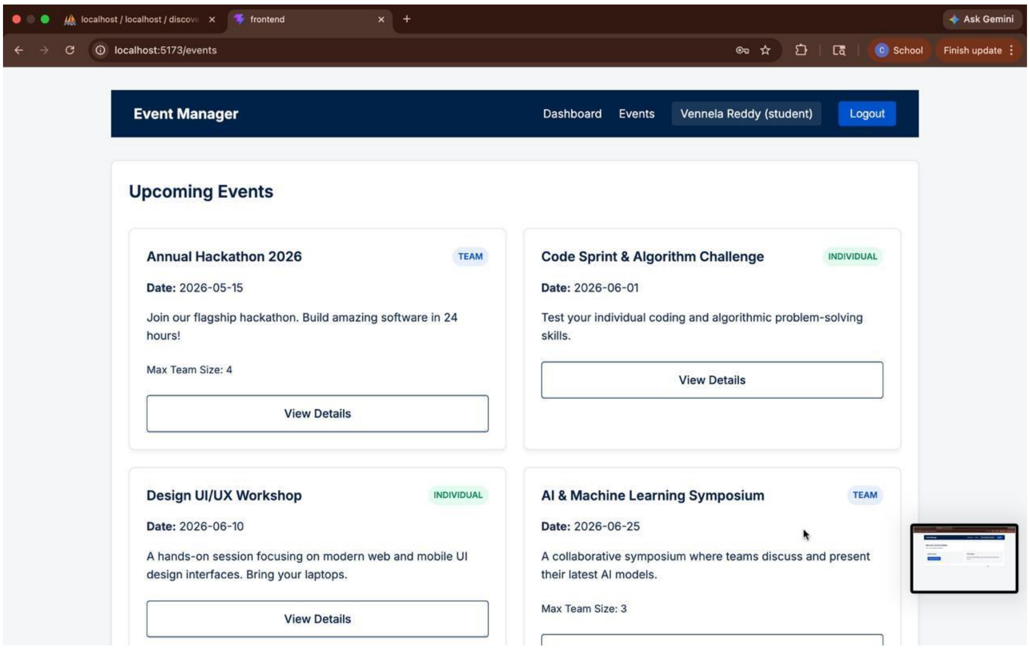

Figure 5.2: EventDashboard

The Events page displays a list of upcoming campus events in a clear and organized for- mat. Each event is presented as a card containing essential details such as event title, date, description, and participation type (team or individual). For team-based events, the maximum team size is also shown.The interface allows users to easily browse through multiple events and understand their details at a glance. Each event card includes a “View Details” button, which enables users to access more information and proceed with registration.The layout is user-friendly and visually structured, helping users quickly explore available opportunities and choose events based on their interests. Overall, the page simplifies event discovery and improves user engagement within the platform.

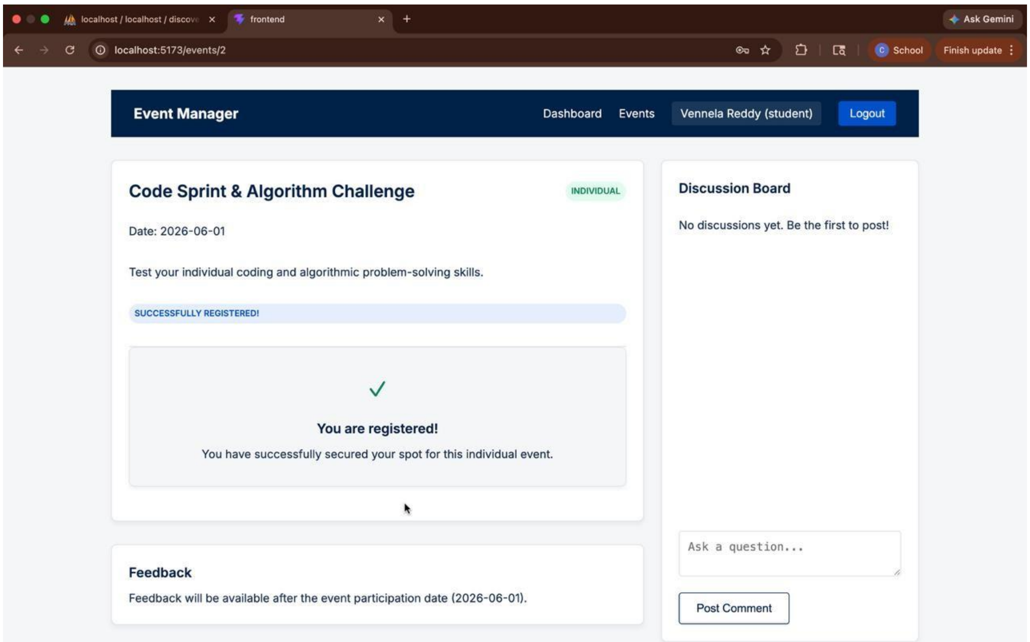

Figure 5.3: RegisterationPage

The Event Details page provides complete information about a selected event, including the event title, date, description, and participation type. It allows users to clearly understand the event before taking any action.Once a user registers for the event, a confirmation message is displayed in- dicating successful registration. This ensures that users receive immediate feedback and confirmation of their participation. The interface also shows a status section to inform users about their registration progress.Additionally, a Discussion Board is included where users can post questions or interact with other participants. A feedback section is also provided, which becomes active after the event date, allowing users to share their experience.Overall, this page enhances user interaction by combining event information, registration status, communication, and feedback features in a single interface.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Platform successfully addresses the challenges associated with traditional event management systems in educational institutions. By providing a centralized and automated solution, the platform simplifies event creation, registration, attendance tracking, and certificate generation The system improves efficiency by reducing manual work, minimizing errors, and enhancing communication between administrators and students. The integration of secure authentication using JWT, along with features like email notifications and PDF certificate generation, ensures a reliable and user-friendly experience.Overall, the project demonstrates how modern web technologies can be effectively used to build a scalable and efficient event management system tailored for campus environments.

6.2 Future Enhancements

Although the system meets current requirements, several improvements can be made in the future to enhance functionality and user experience:

Development of a mobile application for Android/iOS platforms Integration of QR code-based attendance tracking for faster check- ins Implementation of real-time notifications using push services Advanced analytics and reporting dashboard for administrators Integration with third-party platforms such as Google Calendar Support for online/virtual events with live streaming features Multi-language support for better accessibility

Chapter 7
ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

Table 7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of knowledge	Requires in depth engineering knowledge	WK3,WK4, WK5, WK6	Requires knowledge in Spring Boot, MVC architecture, database design, REST APIs, and frontend technologies (Thymeleaf, HTML/CSS) for developing a scalable event management system.
WP2	Conflicting requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances usability and security by ensuring easy student access while maintaining secure admin authentication and data validation.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Involves system design, database schema creation, validation handling, and efficient event filtering and registration mechanisms.
WP4	Familiarity	Partially new problems	WK4, WK8	Integrating real-time event updates, user interaction, and admin analytics introduces challenges beyond traditional manual systems.
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom implementation for event filtering, registration tracking, and statistics generation not fully covered by standard templates.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Addresses requirements of students, faculty coordinators, and administrators ensuring smooth event organization and participation.
WP7	Interdependence	System level integration	WK3, WK5, WK6	Integrates frontend UI, backend services, database, validation, and security modules into a unified system.

The project is analyzed based on Complex Engineering Problem (CEP) attributes such as depth of knowledge, conflicting requirements, and system integration. It requires strong technical knowledge in web development and database systems. The project also handles real-world challenges like security, usability, and multiple stakeholders. These factors confirm that the project qualifies as a complex engineering problem.

Table 7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Spring Boot, Spring MVC, Spring Data JPA, Thymeleaf, HTML/CSS, and relational databases for full-stack development.
EA2	Level of Interactions	Complex interactions between modules	Manages interactions between event management, user registration, validation, authentication, and reporting modules.
EA3	Innovation	Application of modern technologies	Implements a web-based centralized platform with filtering, real-time updates, and secure admin operations.
EA4	Consequences to Society and Environment	Impact on users and society	Enhances student engagement and improves institutional efficiency while reducing manual errors and delays.
EA5	Familiarity	Beyond standard practices	Combines web development, security, and data analytics into a unified system, extending beyond traditional event handling methods.

The project is mapped to Complex Engineering Activities (CEA) including use of diverse technologies and complex module interactions. It demonstrates innovation through a centralized web -based system. The project improves efficiency and user experience while reducing manual work. Hence, it satisfies the characteristics of complex engineering activities

Table 7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Facilitates student participation in academic events, workshops, and seminars, enhancing learning beyond the classroom.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in educational institutions through a centralized event management system.
SDG 11	Sustainable Cities and Communities	Supports smart campus initiatives by digitizing event processes and improving resource management.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent event handling through authentication and controlled access.
SDG 12	Responsible Consumption and Production	Reduces paper usage by enabling online event registration and management

The project is aligned with Sustainable Development Goals (SDGs) such as Quality Education and Industry Innovation. It supports digital transformation in educational institutions. The system reduces paper usage and improves accessibility. Thus, the project contributes positively towards sustainable development.

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SOURCE CODE

```
 FRONTEND (React)
import axios from 'axios';
const api = axios.create({
  baseURL: 'http://localhost/backend/api'
});

import React, { useState } from 'react';

function Login() {
  const [email, setEmail] = useState('');
  const [pass, setPass] = useState('');

  const handleSubmit = async (e) => {
    e.preventDefault();
    const res = await api.post('/auth/login.php', {
      email,
      password: pass
    });
    if(res.status === 200){
      window.location.href = "/dashboard";
    }
  };
}

// BACKEND (PHP)
<?php
$conn = new PDO("mysql:host=localhost;dbname=event_management","root","")
    ;

header("Content-Type: application/json");

if($_SERVER['REQUEST_METHOD']=='POST'){
```

```
  $data = json_decode(file_get_contents("php://input"));

  $stmt = $conn->prepare("SELECT * FROM users WHERE email=?");
  $stmt->execute([$data->email]);

  $user = $stmt->fetch(PDO::FETCH_ASSOC);

  if($user && password_verify($data->password,$user['password'])){
    echo json_encode(["message"=>"Success"]);
  } else {
    echo json_encode(["message"=>"Failed"]);
  }
}
?>

// DATABASE (MySQL)
CREATE TABLE users (
 id INT AUTO_INCREMENT PRIMARY KEY,
 name VARCHAR(255),
 email VARCHAR(255),
 password VARCHAR(255),
 role ENUM('student','admin')
);

CREATE TABLE events (
 id INT AUTO_INCREMENT PRIMARY KEY,
 title VARCHAR(255),
 date DATE
);
```